

Count Data and Models

June 2025

What does it mean to be count data?

- Constrained to be positive
 - Zero or greater
- Discrete
 - Whole number counts
- Rate
 - Counts per unit

Lizard Data Set

Does vegetation or temperature explain the number of sand lizards found at a site?

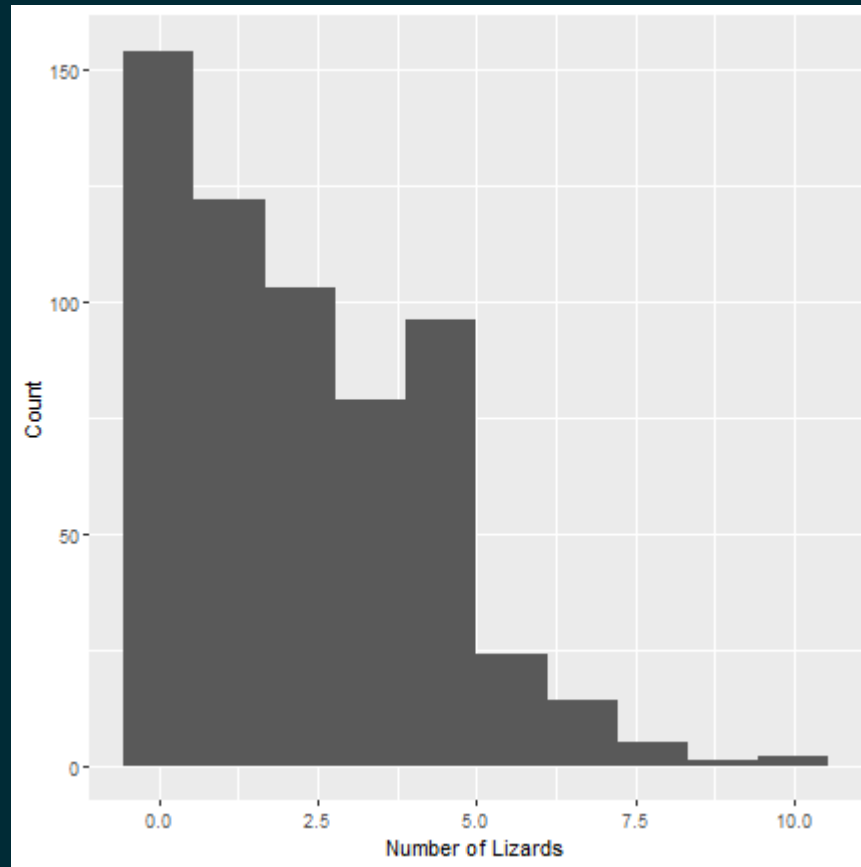
`lizard ~ veg + temp + visit`

- `lizard`: the number of sand lizards found at one of 200 sites
- `veg`: A measure of vegetation between -1 and 1, where 0 is the average vegetation density
- `temp`: A measure of temperature between -1 and 1, where 0 is the average temperature
- `visit`: Takes the value 1, 2 or 3 depending on which visit to the site is being recorded

Data courtesy of Marc Kery

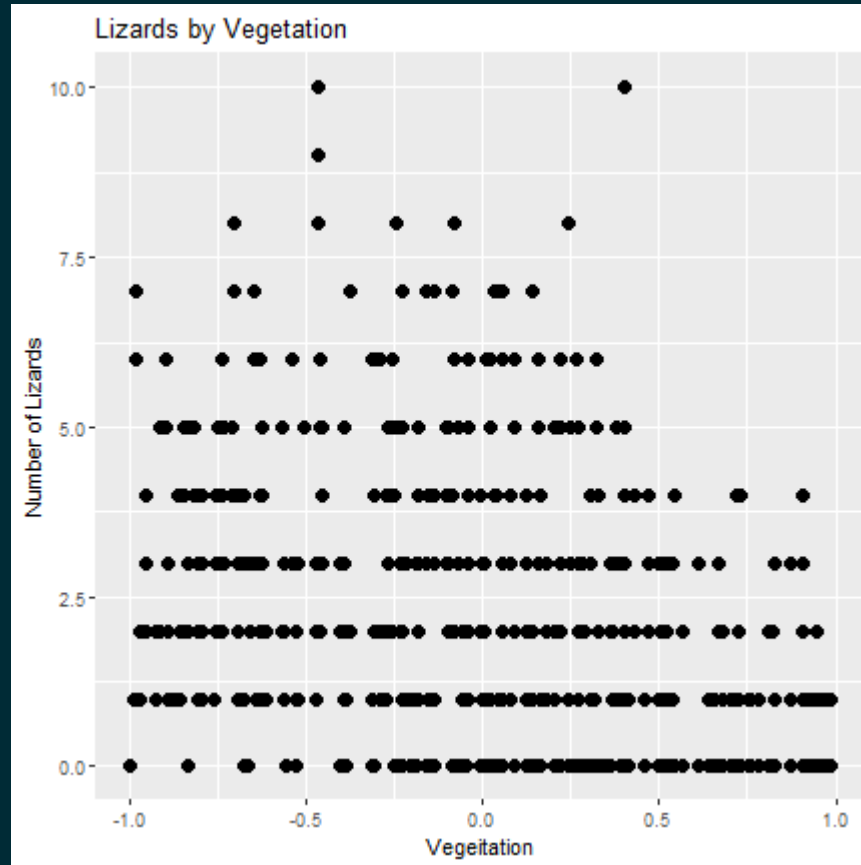
Visualize

```
ggplot(lizard, aes(x=lizard)) +  
  geom_histogram(bins=10) +  
  labs(x="Number of Lizards",  
       y="Count")
```

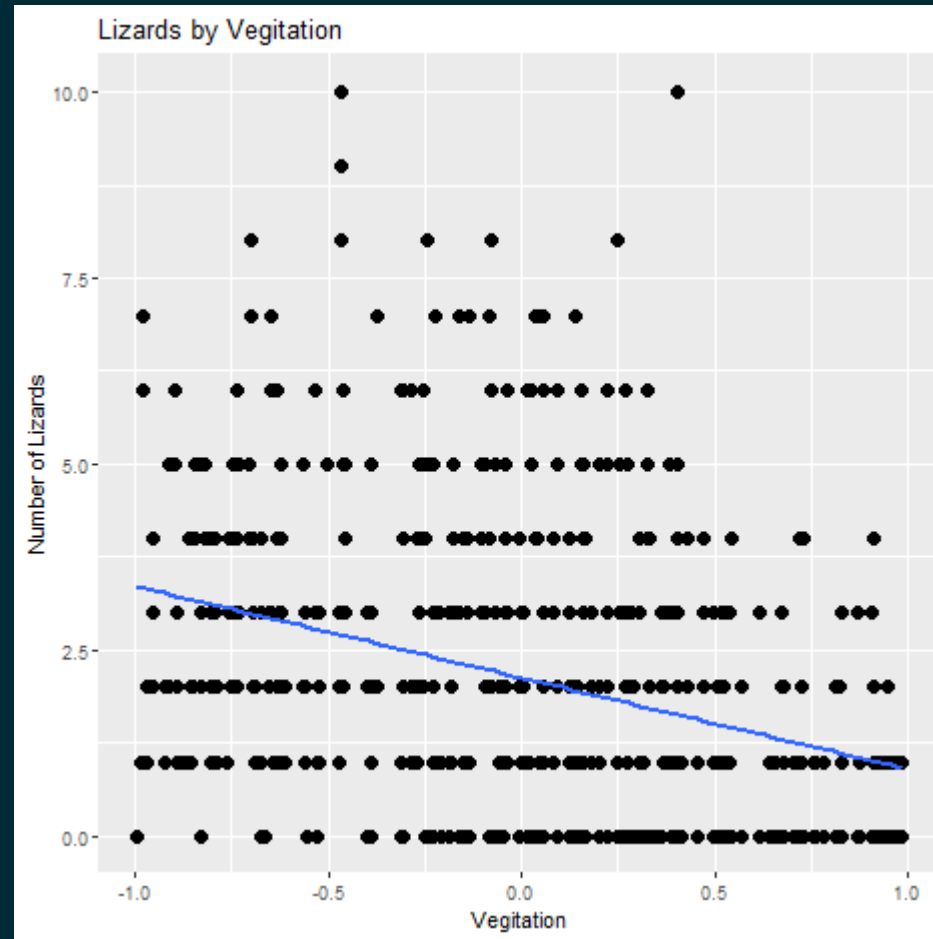


Lizards and Vegetation

```
ggplot(lizard,aes(y=lizard,x=veg))+  
  geom_point(size=3)+  
  labs(title="Lizards by Vegetation",  
        x = "Vegetation",  
        y = "Number of Lizards")
```



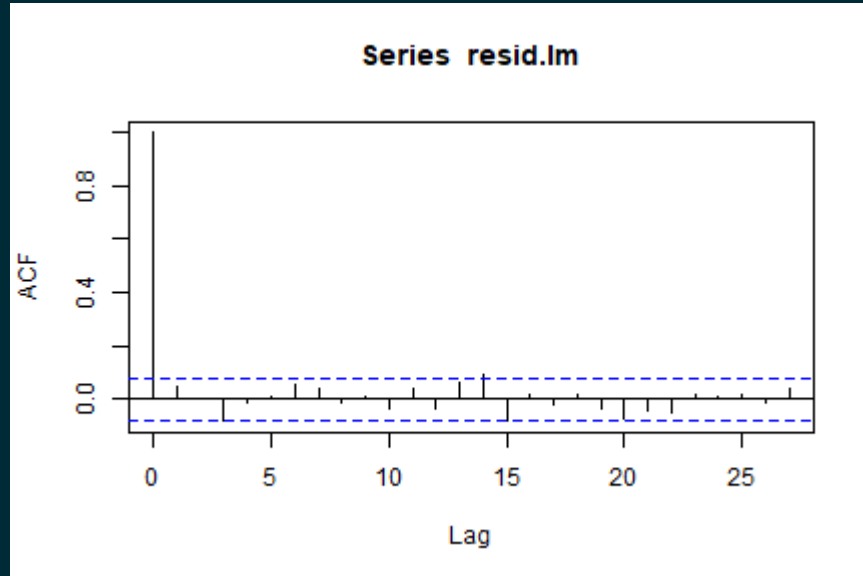
Bad Idea



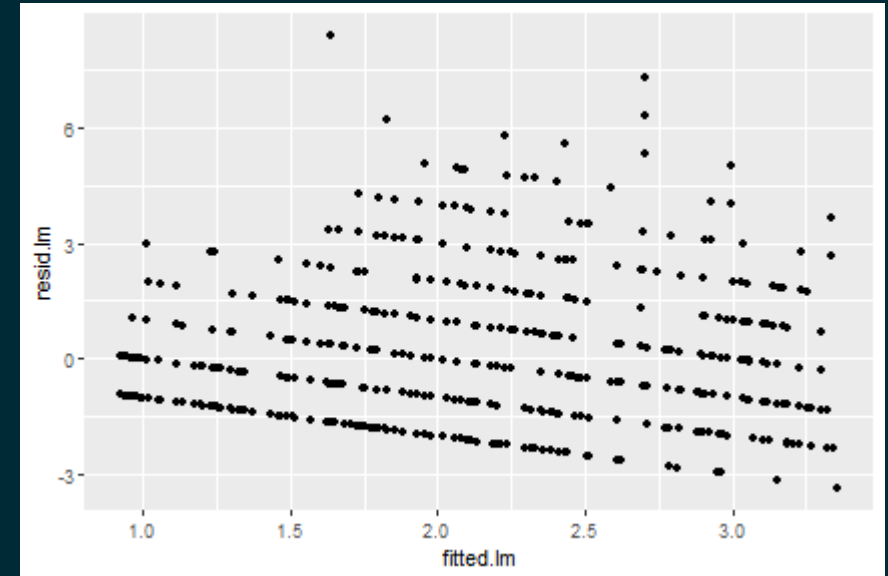
Why bad?

Check Formally

- Independence

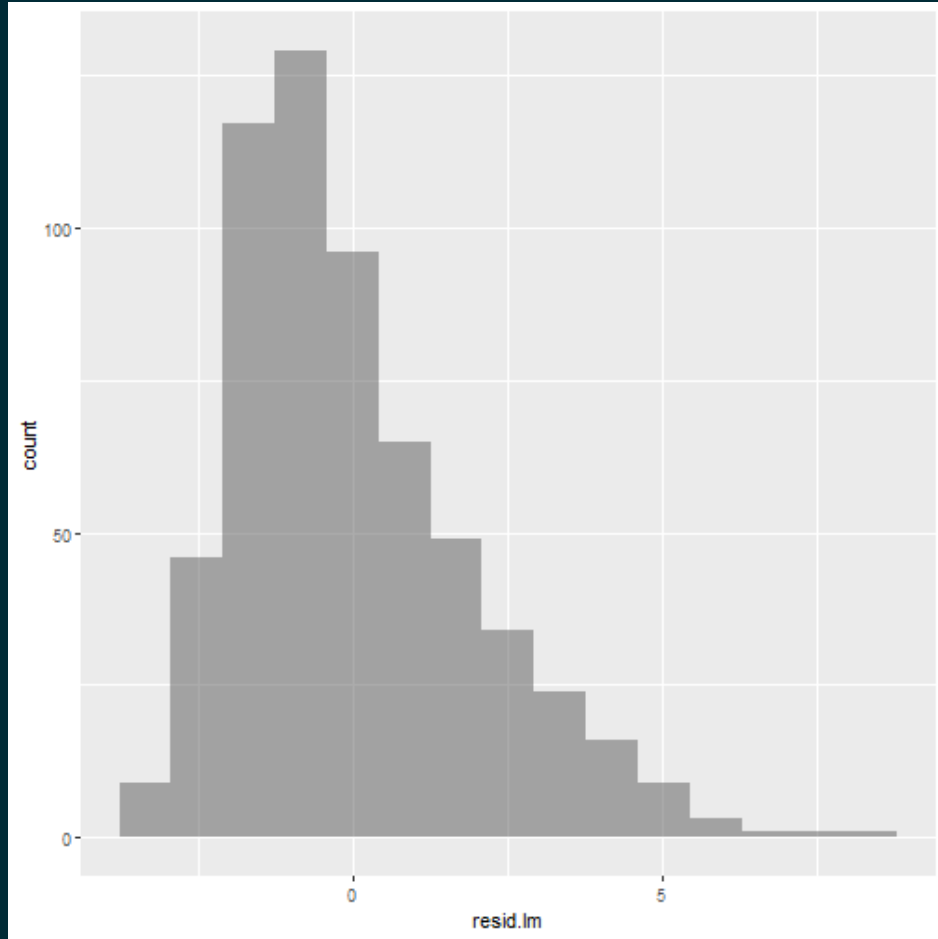


- Error is constant

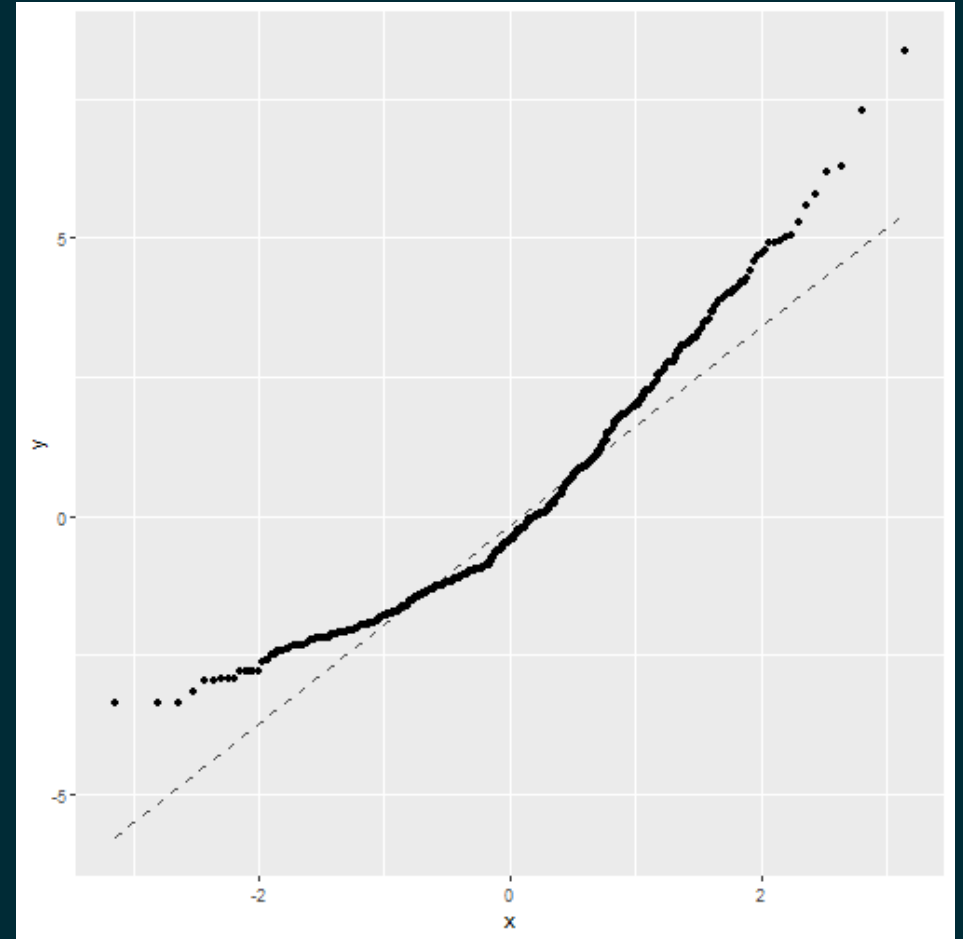


Normality

- Histogram

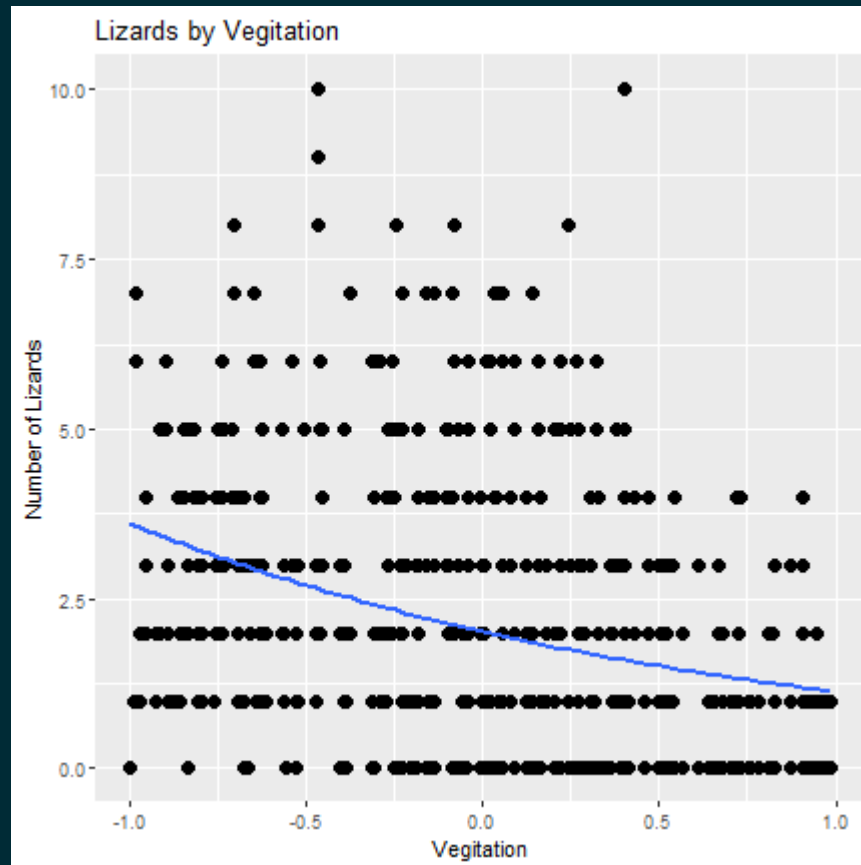


- QQ plot



Good Idea

```
ggplot(lizard,aes(y=lizard,x=veg))+  
  geom_point(size=3) +  
  labs(title="Lizards by Vegetation",  
        x = "Vegetation",  
        y = "Number of Lizards")+  
  geom_smooth(method="glm",method.args = list(family = "poisson"),se=F)
```



How do we get there?

- Generalized linear model (GLM)
- Relax some assumptions
 - Linear on the scale of the linear predictor
 - Residuals are appropriate for the chosen distribution
- Still need independence

Poisson GLM

- Response variable

$$lizard_i \sim Pois(\lambda_i)$$

- Explanatory variable

$$g(\lambda_i) = \eta_i = \beta_0 + \beta_1 veg_i$$

- Link function

$$g(\lambda_i) = \log(\lambda_i) = \eta_i = \beta_0 + \beta_1 veg_i$$

- Invertible

$$\lambda_i = g^{-1}(\eta_i) = \exp(\beta_0 + \beta_1 veg_i)$$

Model Fitting

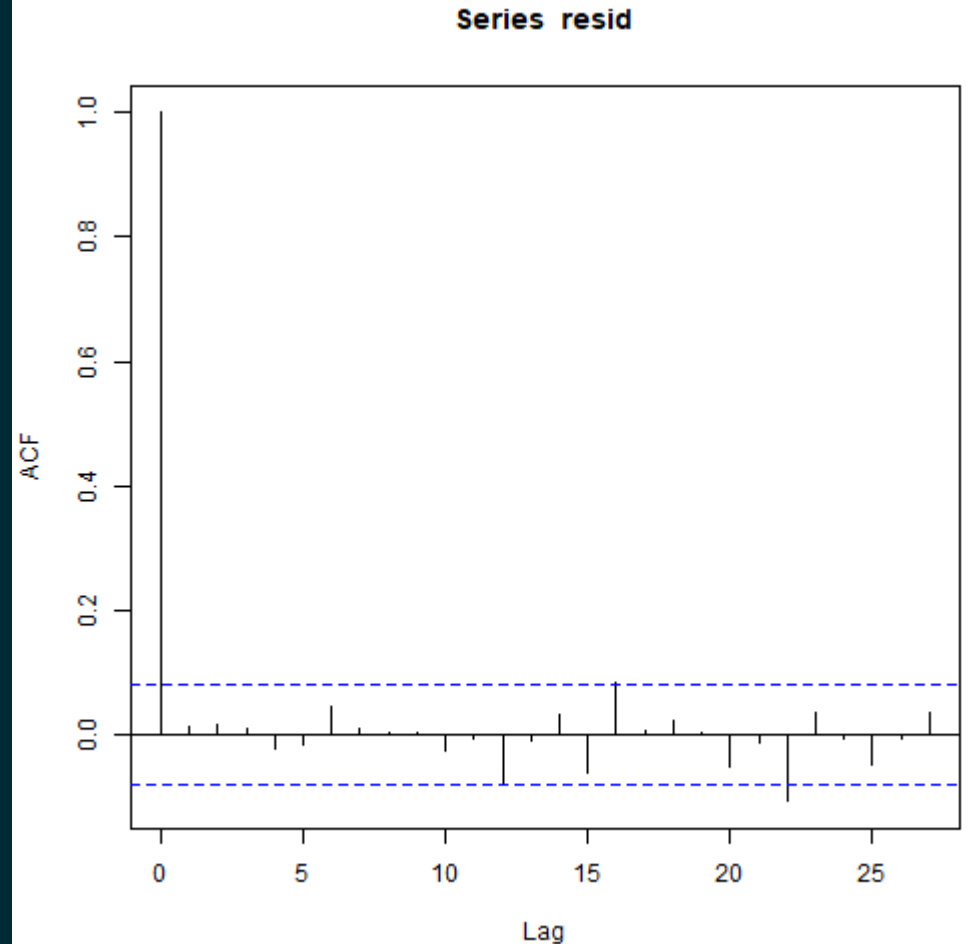
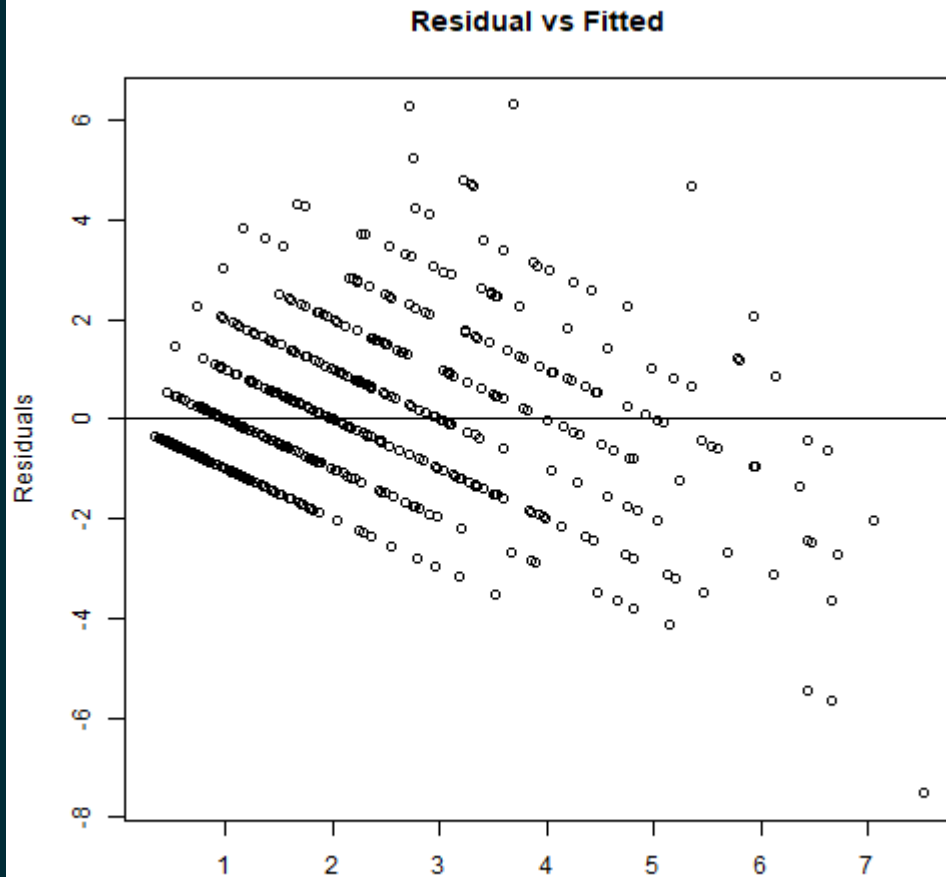
```
pois.mod <- glm(lizard~veg+temp+as.factor(visit),family=poisson(link="log"),data=lizard)
summary(pois.mod)
```

```
##
## Call:
## glm(formula = lizard ~ veg + temp + as.factor(visit), family = poisson(link = "log"),
##      data = lizard)
##
## Coefficients:
##              Estimate Std. Error z value Pr(>|z|)
## (Intercept)    0.44786    0.05370   8.340  <2e-16 ***
## veg           -0.52963    0.05095 -10.396  <2e-16 ***
## temp            1.07834    0.05778  18.664  <2e-16 ***
## as.factor(visit)2 -0.01730    0.06814  -0.254    0.800
## as.factor(visit)3  0.03696    0.06694   0.552    0.581
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## (Dispersion parameter for poisson family taken to be 1)
##
##      Null deviance: 1263.56  on 599  degrees of freedom
## Residual deviance:  731.58  on 595  degrees of freedom
## AIC: 1987.5
##
## Number of Fisher Scoring iterations: 5
```

Assumptions

```
resid <- residuals(pois.mod,type="response")  
fit <- fitted(pois.mod)  
plot(fit,resid,main="Residual vs Fitted",xlab="Fitted",ylab="Residuals")  
abline(h=0)
```

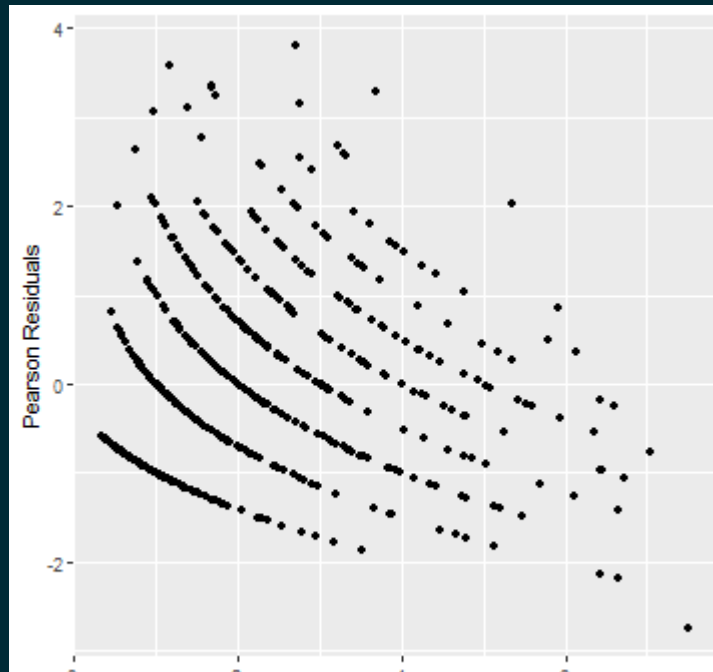
```
acf(resid)
```



Right amount of variance?

- Use Pearson's residuals
 - Scaled by expected variance
 - Approximately constant variance
 - Standard deviation of 1

```
p.resid <- resid(pois.mod, type='pearson')  
gf_point(p.resid ~ fit) |>  
  gf_labs(x='Fitted Values', y='Pearson Residuals')
```



Overdispersion

- More variance than expected
 - Poisson distribution: $\mu = \sigma^2$
 - Can check with the quasi-Poisson

```
quasi.mod <- glm(lizard~veg+temp+as.factor(visit),family=quasipoisson(),data=lizard)
summary(quasi.mod)
```

```
##
## Call:
## glm(formula = lizard ~ veg + temp + as.factor(visit), family = quasipoisson(),
##      data = lizard)
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)    0.44786    0.05638   7.944 9.83e-15 ***
## veg            -0.52963    0.05348  -9.902 < 2e-16 ***
## temp           1.07834    0.06066  17.778 < 2e-16 ***
## as.factor(visit)2 -0.01730    0.07153  -0.242  0.809
## as.factor(visit)3  0.03696    0.07028   0.526  0.599
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## (Dispersion parameter for quasipoisson family taken to be 1.102188)
##
##      Null deviance: 1263.56  on 599  degrees of freedom
## Residual deviance:  731.58  on 595  degrees of freedom
## AIC: NA
##
## Number of Fisher Scoring iterations: 5
```


How big is too big?

- overdispersion < 1.5, generally okay
- overdispersion > 15, big problems
 - Alternative is the negative-binomial
- Variance can be greater than the mean
- Still using the log link

```
library(MASS)
nb.mod <- glm.nb(lizard~veg+temp+as.factor(visit), data=lizard)
summary(nb.mod)
```

```
##
## Call:
## glm.nb(formula = lizard ~ veg + temp + as.factor(visit), data = lizard,
##       init.theta = 15.23779104, link = log)
##
## Coefficients:
##              Estimate Std. Error z value Pr(>|z|)
## (Intercept)    0.43463    0.05777   7.524 5.31e-14 ***
## veg           -0.56857    0.05545 -10.254 < 2e-16 ***
## temp           1.10126    0.06209  17.737 < 2e-16 ***
## as.factor(visit)2 -0.01445    0.07437  -0.194    0.846
## as.factor(visit)3  0.04144    0.07328   0.565    0.572
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## (Dispersion parameter for Negative Binomial(15.2378) family taken to be 1)
##
## Null deviance: 1124.34  on 599  degrees of freedom
```

Which to use?

- Judgement call
- Can compare

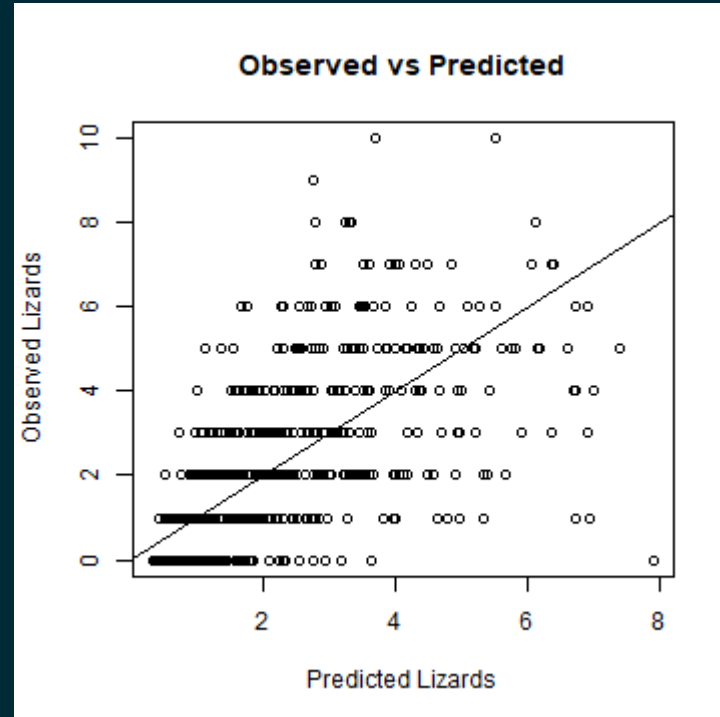
```
library(lmtest)
lrtest(pois.mod,nb.mod)
```

```
## Likelihood ratio test
##
## Model 1: lizard ~ veg + temp + as.factor(visit)
## Model 2: lizard ~ veg + temp + as.factor(visit)
##   #Df  LogLik Df  Chisq Pr(>Chisq)
## 1    5 -988.73
## 2    6 -985.46  1 6.5459    0.01051 *
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

Assessment

What might be a good why to assess model fit?

- R^2 works best for simple linear regression
 - Are some other related options, won't cover here
- Observed versus predicted



Interpretation

How do we interpret the intercept in this case?

- Zero is the average in our data
 - Number of lizards for average vegetation and temperature
 - $\exp(0.434) = 1.54$
- Start with the sign of the slopes
 - Decrease with increasing vegetation
 - Increase with increasing temperature
 - No effect of visit (slopes not statistically different from zero)

More Interpretation

What is the difference between the scale of the linear predictor and scale of the response?

- Exponentiate the slopes
 - Percentage change in response variables for a unit increase in the explanatory variable
 - $\exp(-0.569) = 0.566$
 - $\exp(1.101) = 3.01$
- What is a unit?
- -1 to 0
- 0 to 1
- Scale by raising to the power
 - $\exp(-0.569)^{0.1} = 0.95$
 - $\exp(1.101)^{0.1} = 1.12$